

HIGHER SCHOOL CERTIFICATE EXAMINATION

2024

Mathematics Advanced

Examiner: Sami El Hosri

General Instructions

- Reading Time – 10 minutes
- Working Time – 3 hours
- Write using black pen
- Calculators approved by NESAC may be used
- A separate reference sheet is provided at the back of this paper.
- All necessary working should be shown in Questions 11 – 33.

Total marks – 100

Section I: Multiple Choice

Questions 1 – 10 10 marks

- Attempt all questions
- Allow about 15 minutes for this section.

Section II: Extended Response

Questions 11 – 33 90 marks

- Attempt all questions
- Allow about 2 hours 45 minutes for this section.

Section I

Questions 1 – 10 (10 marks)

Read each question and choose an answer A, B, C or D.

Allow about 15 minutes for this section.

Use the multiple-choice answer sheet for Questions 1-10.

1 Find $\int (e^3 + x^2) dx$.

A) $\frac{e^4}{4} + \frac{x^3}{3} + c$

B) $3e^2 + 2x + c$

C) $e^3x + \frac{x^3}{3} + c$

D) $e^3 + \frac{x^3}{3} + c$

2 The parabola $f(x) = -x^2 + 9mx + 2$ has a maximum at $x = \frac{2}{m}$, where m is a constant such that $m \neq 0$.

What are the values of m ?

A) $m = \frac{2}{3}$ or $m = -\frac{2}{3}$

B) $m = \frac{4}{9}$ or $m = -\frac{4}{9}$

C) $m = \frac{9}{4}$ or $m = -\frac{9}{4}$

D) $m = \frac{3}{2}$ or $m = -\frac{3}{2}$

- 3 In a game of chance a player is to spin a spinner with sectors of unequal sizes.

The following table gives the probability distribution of the money that the player could win.

Money prize	\$100	\$400	\$500	\$800
Probability	$\frac{1}{4}$	$\frac{1}{8}$	x	y

The expected money prize to be won in a game is \$400.

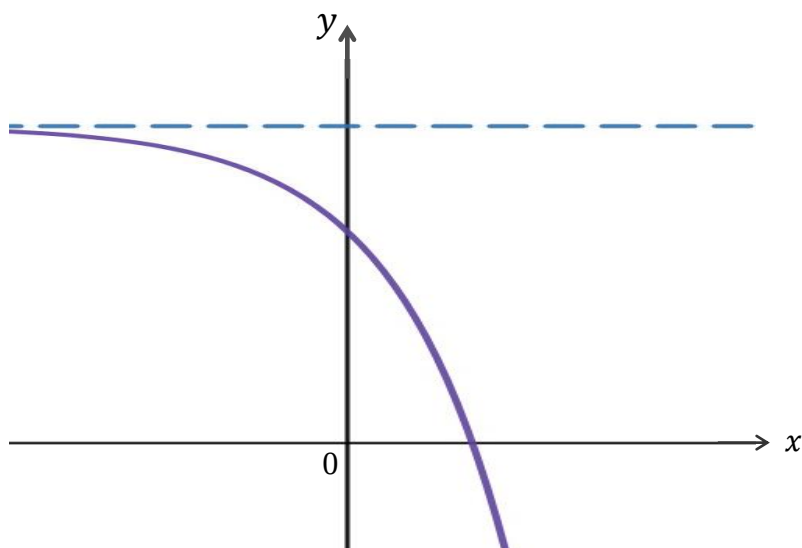
What is the probability of winning \$500?

- A) $\frac{1}{24}$
- B) $\frac{7}{12}$
- C) $\frac{1}{6}$
- D) $\frac{5}{8}$
- 4 What is the domain of $y = \sqrt{x} + \ln(1 - x)$?
- A) $[0, 1]$
- B) $(1, +\infty)$
- C) $(0, +\infty)$
- D) $[0, 1)$

- 5 The maximum of the function $f(x) = 5r - r \cos px$ is $r^2 + 5$, where r is a positive real number.

Given that the minimum of $f(x)$ is greater than 10, what is the minimum value of $f(x)$?

- A) 15
- B) 20
- C) 25
- D) 30
- 6 The graph of the function $y = f(x)$ is as shown below.

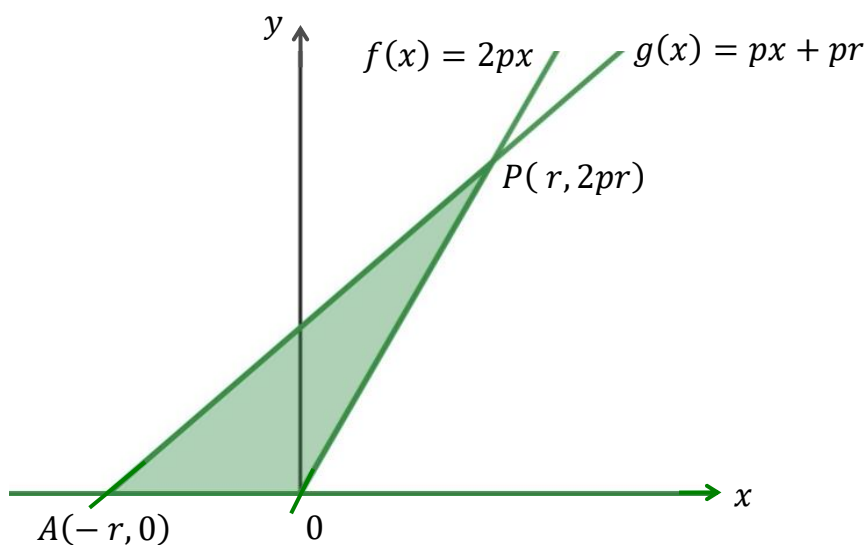


Which row in the table could correctly represent the equation of the function $y = f(x)$?

	Equation	Value of p	Value of q	Value of r
A)	$y = -p + r^{-qx}$	$p > 0$	$q > 0$	$0 < r < 1$
B)	$y = p - r^{qx}$	$p > 0$	$q < 0$	$r > 1$
C)	$y = -p - r^{-qx}$	$p < 0$	$q < 0$	$r > 1$
D)	$y = p - r^{-qx}$	$p < 0$	$q > 0$	$0 < r < 1$

- 7 A class of 28 students sat for an English test.
- The following are the five number summary for the students' results in that test.
- The lowest mark is 68.
- The lower quartile is 72.
- The median is 74.
- The upper quartile is 80.
- The highest mark is 96.
- What is the maximum number of students who could have scored 72?
- A) 13
- B) 14
- C) 12
- D) 7
- 8 Which of the following is an expression for $\log_2 (x \log_{3x} 9x^2)$, where $x > 0$?
- A) x
- B) $\log_2 x$
- C) $2 + \log_2 x$
- D) $1 + \log_2 x$
- 9 Let $f(x)$ and $g(x)$ be two odd functions.
- Which of the following is an odd function?
- A) $f(x) g(x)$
- B) $f(x) - f(g(x))$
- C) $f(x) f(x) + g(x)$
- D) $g(x) g(x) + f(x)$

- 10 Consider the lines $f(x) = 2px$ and $g(x) = px + pr$, where p and r are positive constants.



The shaded triangle OAP bounded by the lines $y = f(x)$, $y = g(x)$ and the x axis is as shown in the diagram.

The area of the triangle OAP is pr^2 .

What is the area of the triangle bounded by $y = f(x - r)$, $y = g(x + r)$ and the x axis?

- A) $9pr^2$
- B) $12pr^2$
- C) $14pr^2$
- D) $15pr^2$

ADVANCED MATHEMATICS

Section II

90 marks

Attempt Questions 11 – 33

Allow about 2 hours and 45 minutes for this section.

INSTRUCTIONS

- Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses.
 - Your responses should include relevant mathematical reasoning and/or calculations.
 - Extra writing space is provided at the back of this booklet.
If you use this space, clearly indicate which question you are answering.
-

Question 11 (2 marks)

Find the coordinates of the centre and the radius of the circle with equation

2

$$x^2 + 6x + y^2 - 8y - 11 = 0$$

Question 12 (3 marks)

Find the equation of the tangent to the curve $y = \tan x$ at the point where $x = \frac{\pi}{4}$.

3

Question 13 (2 marks)

Soccer and Volleyball are two sports students may choose to play.
For a group of 50 students, the following is known.

- 12 students play Soccer only.
- 20 students do not play Soccer.
- 8 play neither.

A student is chosen at random.

2

Find the probability that the student plays only one of the two sports.

Question 14 (4 marks)

A discrete probability distribution is given by

$$P(X = x) = \frac{x^2 - x}{20}, \text{ where } x = 2, 3 \text{ and } 4.$$

a) Show the expected value $E(X) = 3\frac{1}{2}$.

2

b) Calculate the standard deviation, correct to two decimal places.

2

Question 15 (2 marks)

Solve $\log_4 (x - 6) = 2 - \log_4 x$.

2

Question 16 (2 marks)

The manager of a new zoo recorded the number of people that attended the zoo in the first 10 weeks.

The number of people who attended the zoo in the first week was 2000.

In each subsequent week the number of people attending the zoo was 800 more than the previous week.

Find the total number of people who attended the zoo in these 10 weeks.

2

Question 17 (2 marks)

Use the trapezoidal rule with 3 function values to find the area between the curve $y = \ln x$, the x axis, and the lines $x = 1$ and $x = 3$.

2

Give your answer correct to 2 decimal places.

[illegible]

Question 18 (6 marks)

In a year 12 trial only seven of the Chemistry students sat for the examinations of Chemistry and Italian.

The table below shows the marks of these seven students in these two subjects.

Chemistry	42	52	68	80	78	50	48
Italian	74	78	48	58	52	84	76

- a) Complete the table below and use the values found to help find the equation of the regression line (least squares line of best fit) for Chemistry and Italian.

3

	Mean	Standard Deviation
Score in Chemistry	$\bar{x} =$	$\sigma_x =$
Score in Italian	$\bar{y} =$	$\sigma_y =$
The equation of the least squares line of best fit is $y = m x + b$ where $m = r \times \frac{\sigma_y}{\sigma_x}$, r is Pearson's correlation coefficient and $b = \bar{y} - m \bar{x}$		

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- b) Ralph's and Kevin's Chemistry marks were 60 and 90 respectively, but they both did not sit for the Italian trial. Use the regression line to estimate their Italian marks for the trial.

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- c) State whether the estimated Italian mark for Kevin, obtained in part (b), is reliable. Give reasons for your answer.

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Question 19 (2 marks)

The derivative function of a curve is $f'(t) = 2e^{2t} + 4e^{4t}$.

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The curve passes through the origin.

Find the equation of the curve $y = f(t)$.

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Question 20 (4 marks)

David's scored 83 in a test which had a mean of 65 and a standard deviation of 12.

The teacher decides to scale the marks by increasing the mean to 70 and decreasing the standard deviation to 10.

a) What is David's new scaled score?

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b) Ronald is a student in David's class and his new score is 90.

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What is Ronald's original score?

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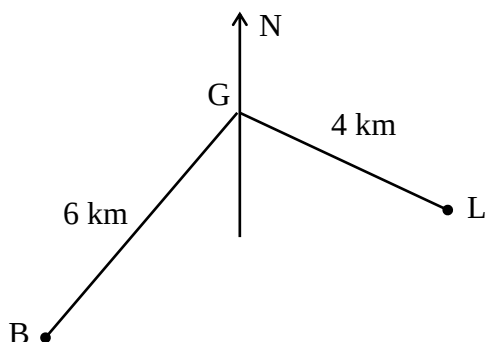
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Question 21 (4 marks)

Layla and Belinda leave the gym G to return to their homes.

Layla cycled 4 km on a bearing 130° to her house at L.

Belinda walked 6 km on a bearing 235° to her home at B.



- a) Find the distance BL correct to 3 decimal places.

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- b) Find the bearing of Layla's house L from Belinda's home at position B, correct to the nearest degree.

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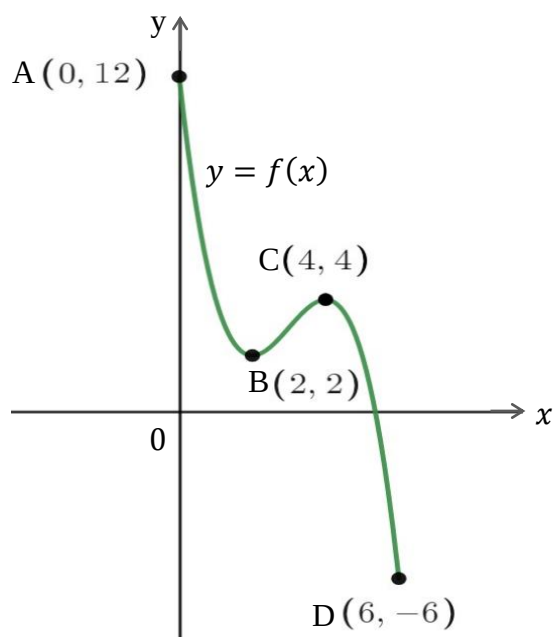
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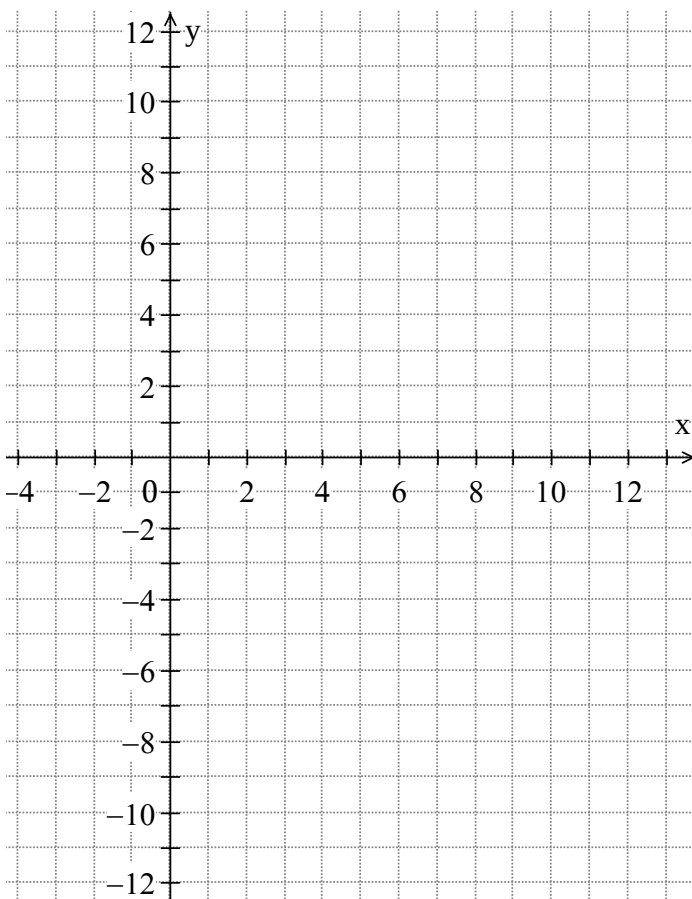
Question 22 (3 marks)

The diagram shows the graph of the function $y = f(x)$.



Sketch the graph of the function $y = 1 - f\left(\frac{x}{2}\right)$ showing the coordinates of the turning points and the y intercepts on the number plane provided below.

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Question 23 (4 marks)

The table below shows the future value of an annuity with a contribution of \$1 at the end of each period.

Interest rate per period							
Period	0.75%	1.00%	1.50%	2.00%	2.50%	3.00%	4.00%
1	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
2	2.0075	2.0100	2.0150	2.0200	2.0250	2.0300	2.0400
3	3.0226	3.0301	3.0452	3.0604	3.0756	3.0909	3.1216
4	4.0452	4.0604	4.0909	4.1216	4.1525	4.1836	4.2465
5	5.0756	5.1010	5.1523	5.2040	5.2563	5.3091	5.4163
6	6.1136	6.1520	6.2296	6.3081	6.3877	6.4684	6.6330
7	7.1595	7.2135	7.3230	7.4343	7.5474	7.6625	7.8983
8	8.2132	8.2857	8.4328	8.5830	8.7361	8.8923	9.2142

Jonah opened an account and deposited \$4 000 at the end of the year and continued to deposit \$4 000 at end of each year until the end of the 3rd year.

At the of 4th year, he deposited \$6000 and continued to deposit \$6 000 at end of each year until the end of the 6th year.

The money in his account was earning interest at 3% per annum compounded annually.

How much money will Jonah have in his account at the end of 7 years?

4

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

Question 24 (6 marks)

The mass of rice bags produced by a company is normally distributed with a mean of 2 400 grams and standard deviation of 10 grams.

The company product manager wants to inspect 2 bags.

- a) Find the probability that the two bags each weigh more than 2 420 grams.

3

- b) Given that both bags weigh more than 2 420 grams, find the probability that both of them will weigh more than 2 430 grams.

3

Question 25 (5 marks)

A continuous random variable X has the following probability density function.

$$f(x) = \begin{cases} \frac{1}{(2x+4)\ln 2} & \text{for } 6 \leq x \leq 30 \\ 0 & \text{for all other values of } x \end{cases}$$

a) Find the cumulative distribution function.

3

b) Find the median.

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Question 26 (3 marks)

The line $y = 5x - c$ is tangent to the parabola $y = 2x^2 - 3x - 1$ at a point $P(a, b)$.
Find the value of c .

3

Question 27 (4 marks)

Consider the geometric series.

$$1 + (-3 + 2^x) + (-3 + 2^x)^2 + (-3 + 2^x)^3 + \dots$$

a) For what values of x does the limiting sum of this series exist?

2

b) If the limiting sum of this series is $\frac{1}{4 - 2\sqrt{2}}$, what is the value of x ?

2

Question 28 (4 marks)

Consider the curve $f(x) = \left(1 - 2 \cos \frac{x}{2}\right)^2$, where $0 \leq x \leq 4\pi$.

Sketch the graph of the curve showing the coordinates of any stationary points and the coordinates of the end points.

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Question 29 (5 marks)

A particle starts moving from the origin along the x axis.

Its velocity $v \text{ m s}^{-1}$ at a time t seconds is given by

$$v = 4 - \frac{t}{\ln 2} + 16 \log_2 \left(\frac{t}{2} + 1 \right)$$

- a) Find the acceleration of the particle at any time t .

2

[illegible]

- b) Show that the maximum velocity in ms^{-1} in of the particle is $52 - \frac{14}{\ln 2}$.

3

[illegible]

Question 30 (4 marks)

E and F are two events and their complementary events are E' and F' respectively.

Given that $P(F/E) = \frac{1}{4}$, $P(E/F) = \frac{2}{3}$ and $P(E' \cap F') = \frac{1}{4}$

Show that E and F are independent events.

4

[illegible]

Question 31 (4 marks)

Rachel decided to set up a trust fund for her granddaughter Melissa.

She deposited \$500 at her first birthday and continued to deposit the same amount on each of her birthdays until and including her 20th birthday.

The money is invested at 4.8% per annum compounded monthly and Melissa will be able to withdraw the money on her 21st birthday.

a) What will be the value of the first deposit on Melissa's 21st birthday?

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b) What will be the total value of the fund on Melissa's 21st birthday?

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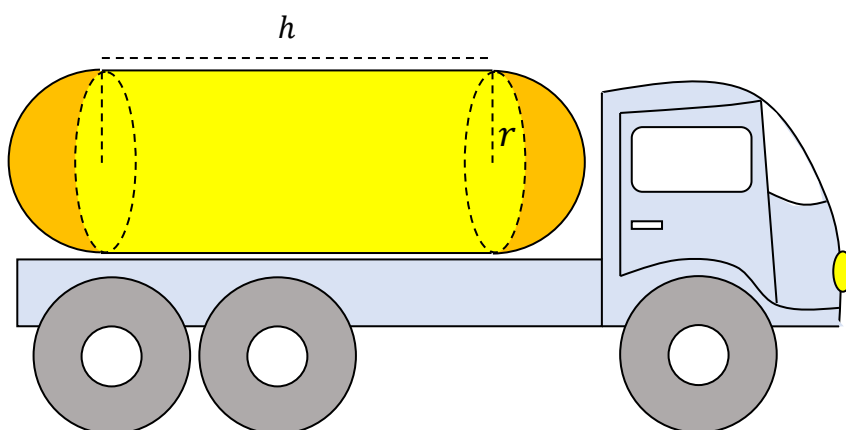
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Question 32 (8 marks)

A transport company is to make a tank to carry liquid chemicals.

The tank is made of a cylinder of radius r and height h metres, and with two hemispheres one at each of the two circular ends. The volume of the tank will be $16p\pi \text{ m}^3$.

The cost of constructing the hemispheres is $\$8p$ per square metre and the cost of constructing the cylinder is $\$3p$ per square metre.



- a) Show the total cost of constructing the tank is $C = 24p\pi \left(r^2 + \frac{4p}{r} \right)$.

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Question 32 continues on page 23

b) Find the value of r in terms of p for which the minimum cost will occur.

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c) Given that the minimum cost occurs when $r = \frac{1}{2}p$, find the minimum cost of constructing the tank.

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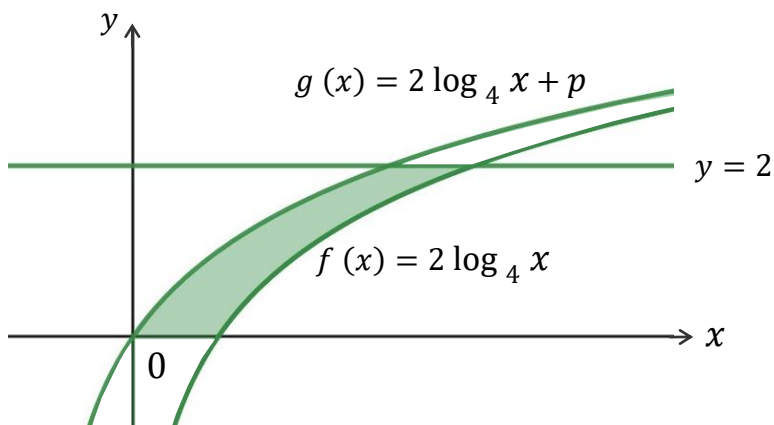
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End of Question 32

Question 33 (7 marks)

Consider the functions $f(x) = 2 \log_4 x$ and $g(x) = p + 2 \log_4 x$, where $x > 0$ and p is a constant.

In the diagram the shaded area bounded by these two functions between $y = 0$ and $y = 2$ is $\frac{3}{2 \ln 2}$.



- a) Show that the value of $p = 1$.

4

[illegible]

Question 33 continues on page 25

b) Consider the functions $r(x) = 2g(x)$ and $h(x) = \frac{k}{f(x)}$, where $x > 0, x \neq 1$ and k is a constant.

3

Find the value of k for which the functions $r(x)$ and $h(x)$ are tangents to each other and the coordinates of their point of contact.

[illegible]

END OF PAPER

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[illegible]

Mathematics Advanced
Mathematics Extension 1
Mathematics Extension 2

REFERENCE SHEET

Measurement

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For $ax^3 + bx^2 + cx + d = 0$:

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

$$\text{and } \alpha\beta\gamma = -\frac{d}{a}$$

Relations

$$(x - h)^2 + (y - k)^2 = r^2$$

Financial Mathematics

$$A = P(1 + r)^n$$

Sequences and series

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1 - r^n)}{1 - r} = \frac{a(r^n - 1)}{r - 1}, r \neq 1$$

$$S = \frac{a}{1 - r}, |r| < 1$$

Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

$$a^x = e^{x \ln a}$$

Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

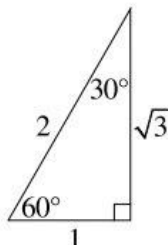
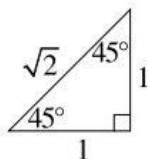
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$



Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \quad \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \quad \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \quad \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

Compound angles

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\text{If } t = \tan \frac{A}{2} \text{ then } \sin A = \frac{2t}{1 + t^2}$$

$$\cos A = \frac{1 - t^2}{1 + t^2}$$

$$\tan A = \frac{2t}{1 - t^2}$$

$$\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$$

$$\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$$

$$\sin A \cos B = \frac{1}{2}[\sin(A + B) + \sin(A - B)]$$

$$\cos A \sin B = \frac{1}{2}[\sin(A + B) - \sin(A - B)]$$

$$\sin^2 nx = \frac{1}{2}(1 - \cos 2nx)$$

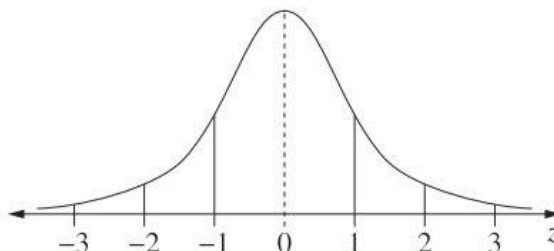
$$\cos^2 nx = \frac{1}{2}(1 + \cos 2nx)$$

Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score
less than $Q_1 - 1.5 \times IQR$
or
more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z -scores between -1 and 1
- approximately 95% of scores have z -scores between -2 and 2
- approximately 99.7% of scores have z -scores between -3 and 3

$$E(X) = \mu$$

$$\operatorname{Var}(X) = E[(X - \mu)^2] = E(X^2) - \mu^2$$

Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) \neq 0$$

Continuous random variables

$$P(X \leq x) = \int_a^x f(x) dx$$

$$P(a < X < b) = \int_a^b f(x) dx$$

Binomial distribution

$$P(X = r) = {}^nC_r p^r (1 - p)^{n-r}$$

$$X \sim \operatorname{Bin}(n, p)$$

$$\Rightarrow P(X = x)$$

$$= \binom{n}{x} p^x (1 - p)^{n-x}, \quad x = 0, 1, \dots, n$$

$$E(X) = np$$

$$\operatorname{Var}(X) = np(1 - p)$$

Differential Calculus

Function

Derivative

$$y = f(x)^n$$

$$\frac{dy}{dx} = n f'(x) [f(x)]^{n-1}$$

$$y = uv$$

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$y = g(u) \text{ where } u = f(x)$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$y = \frac{u}{v}$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$y = \sin f(x)$$

$$\frac{dy}{dx} = f'(x) \cos f(x)$$

$$y = \cos f(x)$$

$$\frac{dy}{dx} = -f'(x) \sin f(x)$$

$$y = \tan f(x)$$

$$\frac{dy}{dx} = f'(x) \sec^2 f(x)$$

$$y = e^{f(x)}$$

$$\frac{dy}{dx} = f'(x) e^{f(x)}$$

$$y = \ln f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

$$y = a^{f(x)}$$

$$\frac{dy}{dx} = (\ln a) f'(x) a^{f(x)}$$

$$y = \log_a f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$$

$$y = \sin^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \cos^{-1} f(x)$$

$$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1 - [f(x)]^2}}$$

$$y = \tan^{-1} f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{1 + [f(x)]^2}$$

Integral Calculus

$$\int f'(x) [f(x)]^n dx = \frac{1}{n+1} [f(x)]^{n+1} + c$$

where $n \neq -1$

$$\int f'(x) \sin f(x) dx = -\cos f(x) + c$$

$$\int f'(x) \cos f(x) dx = \sin f(x) + c$$

$$\int f'(x) \sec^2 f(x) dx = \tan f(x) + c$$

$$\int f'(x) e^{f(x)} dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c$$

$$\int f'(x) a^{f(x)} dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2 - [f(x)]^2}} dx = \sin^{-1} \frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a} + c$$

$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

$$\int_a^b f(x) dx$$

$$\approx \frac{b-a}{2n} \{f(a) + f(b) + 2[f(x_1) + \dots + f(x_{n-1})]\}$$

where $a = x_0$ and $b = x_n$

Combinatorics

$${}^nP_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^nC_r = \frac{n!}{r!(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \cdots + \binom{n}{r}x^{n-r}a^r + \cdots + a^n$$

Vectors

$$|\underline{u}| = |x_1\underline{i} + y_1\underline{j}| = \sqrt{x^2 + y^2}$$

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta = x_1x_2 + y_1y_2,$$

$$\text{where } \underline{u} = x_1\underline{i} + y_1\underline{j}$$

$$\text{and } \underline{v} = x_2\underline{i} + y_2\underline{j}$$

$$\underline{r} = \underline{a} + \lambda \underline{b}$$

Complex Numbers

$$\begin{aligned} z = a + ib &= r(\cos \theta + i \sin \theta) \\ &= re^{i\theta} \end{aligned}$$

$$\begin{aligned} [r(\cos \theta + i \sin \theta)]^n &= r^n(\cos n\theta + i \sin n\theta) \\ &= r^n e^{in\theta} \end{aligned}$$

Mechanics

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v \frac{dv}{dx} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

$$x = a \cos(nt + \alpha) + c$$

$$x = a \sin(nt + \alpha) + c$$

$$\ddot{x} = -n^2(x - c)$$